



COMP4141 Slides Week 10

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ENGINEERING

Admin

Sample Exam

- Has been released
- Solutions will be provided later
- Feel free to discuss with me :)

Feedback From Marking

- Make sure to mention the reduction is polynomial time when trying to prove NP-hardness.
- Be clear on what your verifier is when you try to use one.
- Make sure to explain why your algorithm/verifier is polynomial time in NP inclusion proofs.

W9 P5

Problem 5

Consider the following decision problem

2-SAT

Input: A boolean formula φ in 2-CNF (at most 2 literals per clause)

Question: Is φ satisfiable?

(Same as $\neg a \rightarrow b$ and $\neg b \rightarrow a$) $\neg a \rightarrow b$ and $\neg b \rightarrow a$
 $(a \vee b)$ has an edge $\neg b \rightarrow a$

a) Show $2\text{-SAT} \in P$ b) Show $2\text{-SAT} \in NL$

- Last week I made a mistake; the condition for non-satisfiability is that there is a path from $\neg y$ to y and from y to $\neg y$ (i.e. bidirectional reachability).
- The same idea, using that NL is closed under complement, shows that we can find such paths for an unsatisfiable formula.
- If you know more graph theory, this bidirectional reachability means that y and $\neg y$ lie in the same strongly connected component (SCC).

$co NL = NL$

$$a \rightarrow b \rightarrow c \rightarrow \neg a$$

if a is true
 $\Rightarrow \neg a$ is true

but it doesn't tell us anything
if a is false

$$\neg a \rightarrow d \rightarrow e \rightarrow a$$

but with this path, this says a
is true

PP

Probabilistic Algorithms

Probabilistic Algorithms

A probabilistic TM (PTM) M is a type of NTM in which each non-deterministic step is called a *coin-flip step* and has two legal next moves.

The probability of a branch b of M 's computation on input w is

$$\Pr[b] = \frac{1}{2^k},$$

where k is the number of coin-flip steps that occur on b .

The acceptance probability is the sum of the probabilities of all accepting branches (the rejection probability is its complement).

PP

PP

Say that a PTM M probabilistically recognises A if it runs in polynomial time and:

1. $w \in A \implies \Pr[M \text{ accepts } w] > \frac{1}{2}$,
2. $w \notin A \implies \Pr[M \text{ rejects } w] > \frac{1}{2}$

Define **PP** as the class of languages that are probabilistically recognised by polynomial-time PTMs.

Working with PP

$$w \in \text{SAT}$$

$$P_i[w \text{ is accepted}] \geq \frac{1}{2} - \frac{1}{2^{n+1}}$$

For example, SAT $\in$ **PP**.

$$4\left(\frac{1}{2} + \frac{1}{2^{n+1}}\right) \cdot \frac{1}{2^n}$$

PP in the hierarchy

NP $\subseteq$ **PP** $\subseteq$ PSPACE

$$= \frac{1}{2} + \frac{1}{2^{n+2}} \geq \frac{1}{2}$$

$$1. \quad \frac{1}{2} - \frac{1}{2^{n+1}} \text{ Accept}$$

2. Take a random assignment π and accept iff that assignment is satisfying.

If $w \notin A$

$$P_1(w \text{ is rejected}) = \frac{1}{2} \left(1 + \frac{1}{2^{n+1}} \right)$$

Bounded-error Probabilistic Polynomial Time

BPP

→ Bounded / regardless of the input size n , there is a lower bound for correctness

BPP

For $\varepsilon \in [0, \frac{1}{2})$, a PTM M recognises A with error probability ε if:

1. $w \in A \implies \Pr[M \text{ accepts } w] \geq 1 - \varepsilon$,
2. $w \notin A \implies \Pr[M \text{ rejects } w] \geq 1 - \varepsilon$.

BPP is the class of languages recognised by polynomial-time PTMs with error probability $\frac{1}{3}$.

Right $2^{\frac{2}{3}}$ of the time

BPP example

$\text{PRIMES} = \{n \mid n \text{ is prime (in binary)}\}$

$\text{PRIMES} \in \mathbf{BPP}$

Amplification Lemma

Amplification Lemma

Let $\varepsilon \in [0, \frac{1}{2})$ and let $p(n) > 0$ be a polynomial. If a polynomial-time PTM M recognises A with error probability ε , then there exists another polynomial-time PTM M' that also recognises A with error probability $\exp(-p(n))$.

Interpretation

The definition of **BPP** is robust with respect to the choice of error probability; $\frac{1}{3}$ is just one convenient choice in $[0, \frac{1}{2})$.

$\Rightarrow$ if $C \in \text{BPP}$ then $\forall \varepsilon \in [0, \frac{1}{2})$

then is a PTM^(m) such that

$$P[m' \text{ is correct}] \geq 1 - p$$

RP & ZPP

RP

RP

RP is the class of languages recognisable by probabilistic polynomial-time TMs such that:

- If $w \in A$, then $\Pr[M \text{ accepts } w] \geq \frac{1}{2}$,
- If $w \notin A$, then $\Pr[M \text{ rejects } w] = 1$.

(Also called Monte Carlo PTIME.)

$$\mathbf{coRP} = \{A \mid \overline{A} \in \mathbf{RP}\}, \quad \mathbf{ZPP} = \mathbf{RP} \cap \mathbf{coRP}.$$

Results in **RP**

Lemma

$\text{RP} \subseteq \text{BPP}$

$\text{RP} \subseteq \text{NP}$

Las Vegas Algorithms

Las Vegas

A probabilistic TM M is called *Las Vegas* if for all $w \in \Sigma^*$:

$$\Pr(M \text{ accepts } w) \in \{0, 1\},$$

and for some polynomial p , the expected running time of M on inputs of length n is at most $p(n)$.

Las Vegas and ZPP

ZPP is the class of languages recognised by Las Vegas algorithms.

ZPP Problem

ZPP characterisation

Show that **ZPP** can be characterised as the set of languages L such that there exists a probabilistic Turing machine which:

- Runs in polynomial time and halts in ACCEPT, REJECT, or DON'T KNOW,
- On input $x \in L$, halts in ACCEPT with probability $p > \frac{1}{2}$ or in DON'T KNOW,
- On input $x \notin L$, halts in REJECT with probability $p' > \frac{1}{2}$ or in DON'T KNOW.

ZPP Solution

If $L \in \mathbf{ZPP} = \mathbf{RP} \cap \mathbf{coRP}$, do the following on input w :

- Run w on the **RP** machine. If it accepts, then ACCEPT,
- Otherwise run w on the **coRP** machine. If it rejects, then REJECT,
- Otherwise output DON'T KNOW.

Conversely, suppose we have such a machine:

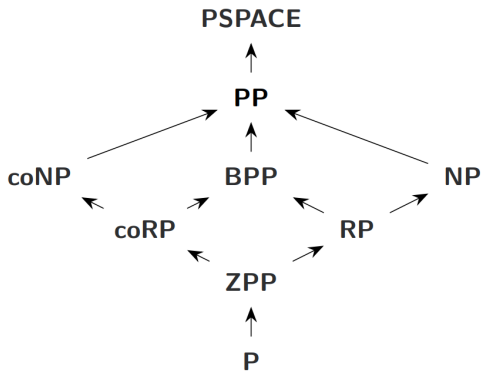
- If we output REJECT on DON'T KNOW, we obtain an **RP** algorithm,
- If we output ACCEPT on DON'T KNOW, we obtain a **coRP** algorithm.

Alternatively, to obtain a Las Vegas algorithm:

- On output DON'T KNOW, rerun the machine.

This yields finite expected running time.

Relationships



W10 P1

Problem 1

Probabilistic computers use a fair coin-flip to make probabilistic choices, is this enough to cover any probability distribution?

(a) Show how a fair coin can simulate rolling a dice. *talked how branching is done*

↓
2 outcomes

↓ ~~this way~~
want 6 outcomes
equally likely

$\Rightarrow$ coin flips give us
to whatever confidence
we want, a probability

Try to flip 3 coins, if we get

- | | | |
|----|-----|------------------------------|
| 1. | HHH | take outcome as 1 (return 1) |
| 2. | HHT | return 2 |
| 3. | HTH | return 3 |
| 4. | HTT | return 4 |
| 5. | THT | return 5 |
| 6. | THT | return 6 |
| 7. | TTT | repeat |
| 8. | TTT | repeat |

← a) probability
 $\frac{1}{6}$

$P_1(\text{After } k \text{ flips that we have not returned})$

$$= \left(\frac{2}{8}\right)^k$$

$$= \frac{1}{4}^k \rightarrow 0 \text{ as } k \rightarrow \infty$$

Problem 1

Probabilistic computers use a fair coin-flip to make probabilistic choices, is this enough to cover any probability distribution?

(b) Show how a fair coin can simulate making a choice with probability p for any $p \in (0, 1)$.

- Return 1 with probability p
- Return 0 otherwise

Try - write p in binary

$$p = 0.p_1 p_2 p_3 \dots$$

$$\left(\begin{aligned} p &= \frac{p_1}{2} + \frac{p_2}{4} + \frac{p_3}{8} + \dots \\ &= \frac{p_1}{2^1} + \frac{p_2}{2^2} + \frac{p_3}{2^3} + \dots \end{aligned} \right)$$

- For i starting at i
(i will count the bit
we are looking at)

- We will flip a coin

$$H \approx 0 \quad T \approx 1$$

What we will do is

if the outcome of coin is
the same then we continue (H and 0
or T and 1)

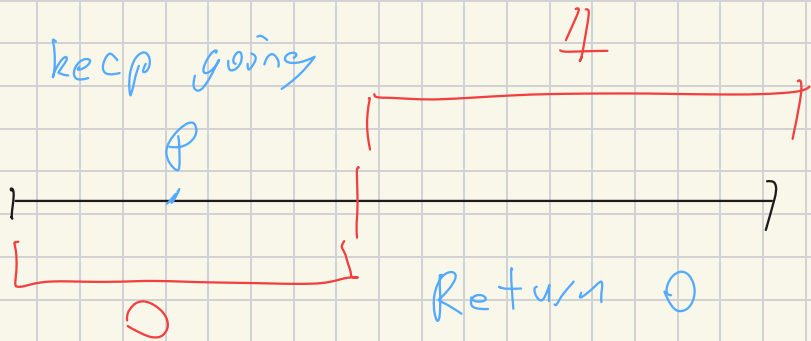
otherwise

- If we roll a $\overline{1}$ ^{≈7} and
the bit is a 0



we will return 0

- If we flip a 1 and the
bit is a 1
we return 1



" Probability that this never terminates is 0

$$\left(\frac{1}{2^k} \rightarrow 0 \text{ as } k \rightarrow \infty \right)$$

$$P_r [\text{Return 1}] = \sum_{\substack{\text{bits in} \\ p \text{ that are 1}}} \frac{1}{2} P_r \left[\text{we get to that bit} \right]$$

$$= \sum_{i: b_i = 1} \frac{1}{2} \left(\frac{1}{2}\right)^{i-1}$$

$$= \sum_{b_i = 1} \left(\frac{1}{2}\right)^i$$

$$= \sum_i \frac{b_i}{2^i}$$

$$= p$$

W10 P2

Problem 2

In the lecture, we showed that $\text{SAT} \in \mathbf{PP}$ and concluded that $\mathbf{NP} \subseteq \mathbf{PP}$. There is actually a missing step in the argument, relating to how $\mathbf{PP}$ interacts with polynomial time reductions. Formulate and prove a result about this, and use it to complete the argument.

Claim: If $B \in \mathbf{PP}$ and $A \leq_p B$ then
 $A \in \mathbf{PP}$

Sol: Since $\forall A \in \mathbf{NP}, A \leq_p \text{SAT}$
and $\text{SAT} \in \mathbf{PP}$, $\mathbf{NP} \subseteq \mathbf{PP}$.



PP is the set of languages (L)
with PTIME PTMs M

$$w \in L \Rightarrow P[M \text{ accepts } w] \geq \frac{1}{2}$$

$$w \notin L \Rightarrow P[M \text{ rejects } w] > \frac{1}{2}$$

(accepts correctly strictly better
than a half for every input)

Proof of claim:

Since $A \leq_p B$, $\exists f$ computable
in PTIME such that

$$w \in A \quad \text{iff} \quad f(w) \in B$$

We can make from
a PTIME PTM that satisfies

* M . Then we show
 $B \in PP$ by M'

$M' =$ " on input w

1. Compute $f(w)$

2. Run $f(w)$ on M''

3. Return same acceptance as M

• M' runs in polynomial time
as $f(w)$ is polynomial in the
of $|w|$, so say $|f(w)| \approx g(w)$
(polynomial g)

and since M runs on word u
in polynomial time $h(|u|)$
(h is another
polynomial)

- M' will run in time $h(g(n))$
which is still $PTIME$

(for step 2)

- We already know step 1 is $PTIME$

Now we just need to argue *.
This is true since

$$\begin{aligned} \text{If } w \in A \Rightarrow f(w) \in B \Rightarrow & \Pr[f(w) \text{ accepted}] \\ & // > \frac{1}{2} \\ & = \Pr[w \text{ accepted}] \end{aligned}$$

and likewise by symmetry $w \notin A$
 $\Rightarrow \Pr[w \text{ rejected}] > \frac{1}{2}$

W10 P3

Problem 3

Prove or disprove: **BPP** is closed under intersection. That is, if $A \in \mathbf{BPP}$ and $B \in \mathbf{BPP}$ then $A \cap B \in \mathbf{BPP}$.

Try (for true): Let M_A, M_B be the PTM
that witness $A, B \in \mathbf{BPP}$
Make M' for $A \cap B$:

$M' = \text{"On input } w$

1. - Run w on M_A
2. - Run w on M_B
- Return true iff M_A, M_B accept

When $w \in A \cap B \rightarrow w \in A, w \in B$

$$\Rightarrow P_i [M_H \text{ accepts}] > \frac{2}{3}$$

$$P_i [M_B \text{ accepts}] > \frac{2}{3}$$

$$P_i [M' \text{ accepts}] \geq \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) = \frac{4}{9}$$

Try again M'_H, M'_B

with

$$P_i [M'_H \text{ is correct}] \geq 1 - \frac{1}{6}$$

$$P_i [M'_B \text{ is correct}] \geq 1 - \frac{1}{6}$$

$$\begin{aligned}\text{If } w \in A \cap B, \quad P_r[M' \text{ accepts}] &\geq \left(\frac{5}{6}\right)^2 \\ &= \frac{25}{36} \\ &\geq \frac{24}{36} \\ &= \frac{2}{3} = 1 - \frac{1}{3}\end{aligned}$$

• If $w \notin A \cap B$ then we want
 $P_r[w \text{ is rejected}] \geq \frac{2}{3}$

If $w \notin A \cap B$ w.l.o.g then $w \notin A$

$$P_r[M'_H \text{ rejects } w] \geq \frac{5}{6}$$

$$P_r[M' \text{ rejects } w]$$

$$\geq P_r[M'_A \text{ rejects}]$$

$$\geq \frac{5}{6} \geq \frac{4}{6} = \frac{2}{3} = 1 - \frac{1}{3} \checkmark$$



W8 P5

Problem 5

Consider the following problem:

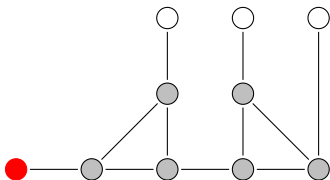
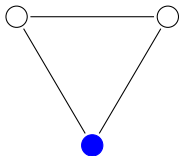
3-COL

Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

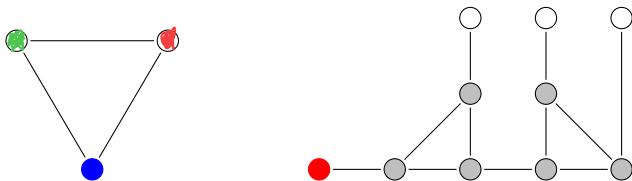
Problem 5

- (a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?



Problem 5

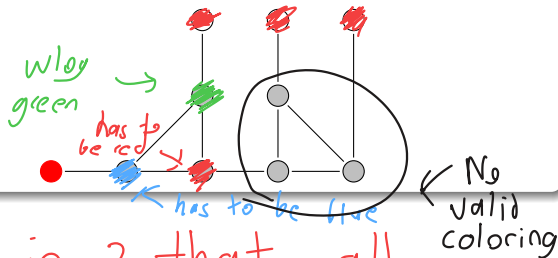
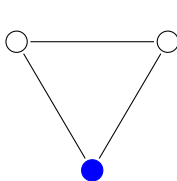
- (a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?



↗
For this one
there is only one
colouring (up to swapping)

Problem 5

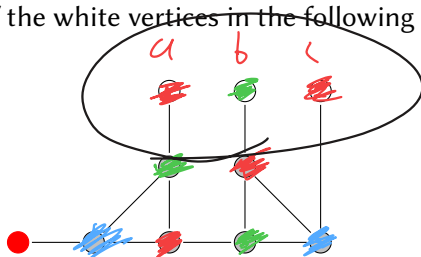
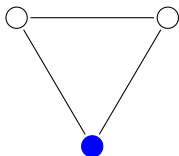
- (a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?



Suppose in 2 that all the white vertices were red

Problem 5

- (a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?



If at least one of a, b, c is green and the others are red then there is a valid colour

Problem 5

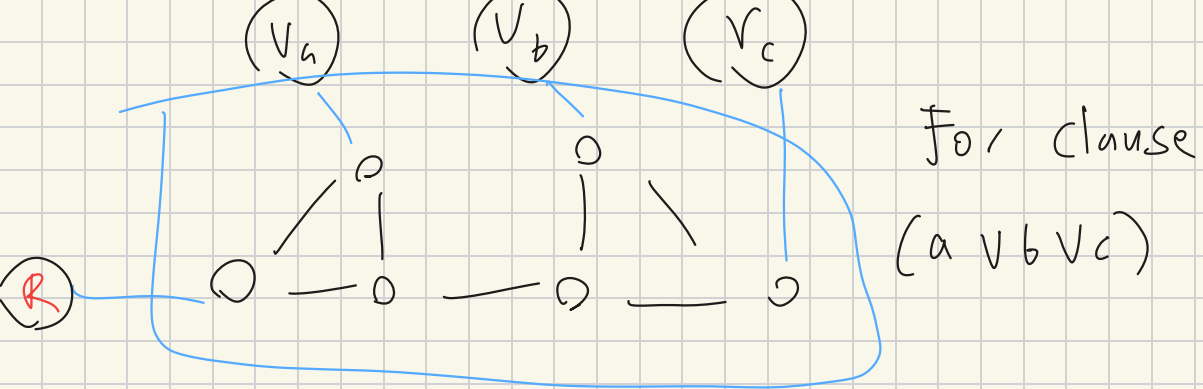
(b) Show that 3-CoL is **NP**-complete, assuming that SAT is **NP**-complete.

- 3-col is in NP (prove it !!!)
- Given boolean formula ϕ
in 3-SAT with n boolean variables
and m clauses

- Make a triangle G, R, B
 3 edges
 3 vertices



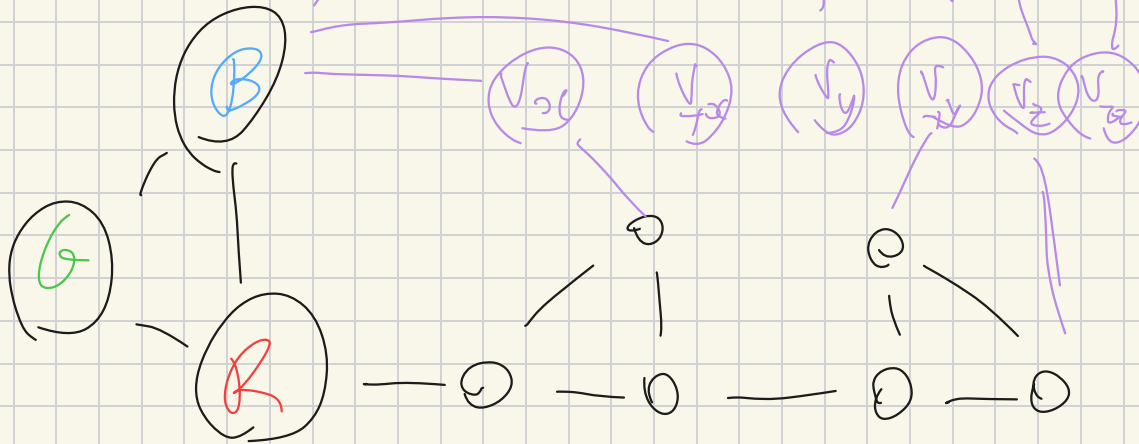
- Make a vertex $\underline{V_{\neg x}}$, $\underline{V_{\neg x}}$
 for each boolean variable x
 and make triangle $V_{\neg x}, V_{\neg x}, B$
 2n edges and
 vertices



Make this gadget
adds 1 inner edge and 6 vertices

In the end G has $O(m+n)$
edges and vertices so this
reduction that outputs G
runs in PTIME.

$$(x \ V \neg y \ V \neg z)$$



$\Gamma_{dp\alpha}$ is

- A valid 3-coloring colors the free "true" V_{α} as green and red otherwise

$\hookrightarrow$, if there at least one green then every clause is satisfied $\Leftrightarrow$ 3 colorable

(need to show both directions in full proof)

W8 P6

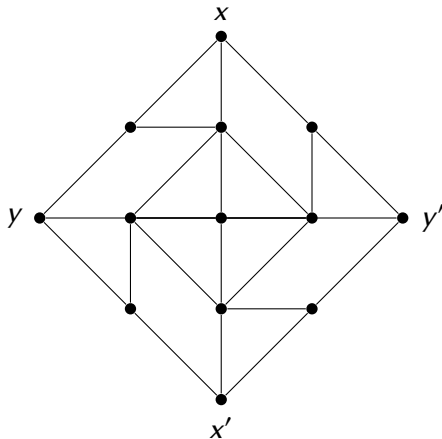
Problem 6

Show that if $\mathbf{P} = \mathbf{NP}$, then every language $A \in \mathbf{P}$ except $A = \emptyset$ and $A = \Sigma^*$, is $\mathbf{NP}$ -complete.

W8 Bonus

Bonus Week 8

Consider the following graph:



Bonus Week 8

(a) Give a valid 3-colouring of this graph where x and y have different colours.

Bonus Week 8

- (a) Give a valid 3-colouring of this graph where x and y have different colours.
- (b) Give a valid 3-colouring of this graph where x and y have the same colour.

Bonus Week 8

- (a) Give a valid 3-colouring of this graph where x and y have different colours.
- (b) Give a valid 3-colouring of this graph where x and y have the same colour.
- (c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

Bonus Week 8

- (a) Give a valid 3-colouring of this graph where x and y have different colours.
- (b) Give a valid 3-colouring of this graph where x and y have the same colour.
- (c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.
- (d) Consider the following decision problem:

PLANAR 3-COL

Input: A planar graph G

Question: Does G have a valid 3-colouring?

Show that PLANAR 3-COL is NP-complete.

Hint: Use the above graph to “uncross” edges.

Bonus Week 8

- (e) Prove that there is no planar graph G , with x, y, x', y' on an outer face (in that order), that satisfies the conditions indicated in (a), (b) and (c), but with a 4-colouring rather than a 3-colouring.